

Chapter Overview

Welcome to Chapter 2 of COSC4036 — Network and System Administration. This chapter covers one of the most critical responsibilities of a system administrator: controlling who has access to a system and precisely what they can do once inside.

Linux is a multi-user operating system — multiple users can be active simultaneously. This power demands careful management. Without proper account administration and access control, one careless or malicious user can compromise the entire system. Every topic in this chapter directly addresses that risk.

Chapter Learning Outcomes

Explain the three Linux user account types: root, human (user), and software accounts.
Distinguish between local and domain user accounts, and between security and distribution groups.
Describe the three Active Directory group scopes: Universal, Global, and Domain Local.
Create, modify, rename, and delete user accounts and groups using Linux commands.
Interpret all seven fields in `/etc/passwd` and all eight fields in `/etc/shadow`.
Explain the difference between local and LDAP account storage.
Use the `finger` and `id` commands to inspect user account information.
Configure password aging using `chage` and `/etc/login.defs`.
Interpret and modify file permissions using both symbolic and octal `chmod` notation.
Change file ownership using `chown` (with dot and colon notation) and `chgrp`.
Configure disk quotas: hard limits, soft limits, and grace periods.
Distinguish between DAC, RBAC, and MAC access control models.